

wasm4pm: First-Mile Mechanical Intelligence

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1 Introduction

The current paradigm of process mining and machine cognition relies heavily on batch-transport architectures. Data is observed at the edge, serialized, transported over network boundaries, and subsequently analyzed in a centralized data warehouse. This creates a structural delay known as "delayed cognition." The *wasm4pm* architecture rejects this paradigm by relocating reasoning and process discovery operators directly to the point of observation via WebAssembly.

By executing within a sovereign WASM sandbox, *wasm4pm* enforces a strict execution boundary. Intelligence is no longer an unbounded pipeline of script execution, but a mathematically bounded, capability-specific transformation. The system compiles down to deterministic binaries that execute first-mile cognition without relying on external system calls or unverified host states.

This monograph presents the formal verification of the *wasm4pm* 115-capability surface (60 algorithms, 55 cognitive breeds). We explicitly downgrade any universal overclaims about AGI or perfectly predictable latency. Instead, we present a testable, falsifiable architecture: bounded inputs yield deterministic, verifiable receipts.

Definition 1.1. First-Mile Execution. The evaluation of an algorithmic or cognitive capability $f : \mathcal{I} \rightarrow \mathcal{O}$ within $O(1)$ network hops from the event source, bounded by memory sandboxing and measured by deterministic computational receipts.

2 Problem: Batch-First Intelligence and Delayed Cognition

The core problem in contemporary process intelligence is the "ETL (Extract, Transform, Load) gap." Organizations observe events at high frequencies, but the algorithmic extraction of insights (such as drift detection, conformance checking, or bottleneck analysis) occurs only after the data has been batched and persisted. This architecture structurally prohibits real-time intervention and autonomic self-healing.

Furthermore, cloud-based intelligence often runs in unbounded runtime environments where memory leaks, infinite loops, and unhandled exceptions are common. A generic Python or Node.js reasoning pipeline cannot mathematically guarantee completion or bound its memory usage without massive external monitoring infrastructure. This lack of bounding makes current intelligent agents unsafe for mission-critical first-mile execution.

wasm4pm solves this by enforcing a capability capability lattice. An operator cannot execute outside its mathematical bounds. The WASM boundary ensures that any attempt to expand state space beyond the allocated heap results in a safe panic, which is caught and transformed into a structural refusal receipt.

Limitation. This approach strictly forbids algorithms that require unbounded data streaming or arbitrary network access. Any cognition requiring global context must be approximated or rejected.

3 First-Mile Mechanical Intelligence

First-Mile Mechanical Intelligence is the thesis that structural, algorithmic, and logical operations can be embedded at the device edge using deterministic WASM binaries. Mechanical intelligence implies that the reasoning process is not a stochastic "black box" but a mechanically verifiable chain of algebraic and geometric operations on discrete event spaces.

In this framework, an event is not admitted to the central ledger until it has been processed and receipted by the relevant cognitive breed or algorithm. For example, an LTL (Linear Temporal Logic) monitor does not wait for a nightly batch run; it observes the event stream

synchronously, verifying the transition against $\sigma, t \models \mathbf{G}\varphi$, and attaching a compliance receipt to the event payload.

Proposition 3.1. *If a capability $\mathcal{C}$ is admitted to the `wasm4pm` kernel, it must terminate in finite time T and emit a cryptographically stable output hash given the same input hash and configuration parameters.*

The 115 capabilities detailed in this monograph represent the baseline mechanical intelligence required to achieve a self-optimizing, autonomous process layer.

4 Law-State Framework

The Law-State framework dictates how events transition from unstructured observation to verified truth. Observations at the edge are inherently untrusted. They must pass through a strict "Court of Admissibility" defined by the capability schema. If an event log violates the structural typing expected by an algorithm, it is refused with a typed `MALFORMED_EVENT_LOG` receipt.

Once admitted, the execution authority is sovereign. The WASM runtime linearly computes the result, completely isolated from host entropy. This guarantees that the execution calculus $f(\text{Binary_Hash}, \text{Input_Hash}, \text{Seed}, \text{Params}) \rightarrow (\text{Output}, \text{Receipt_Hash})$ holds perfectly.

Declared vs. Admitted Capability. A capability is merely declared if its Rust source code exists. It is admitted to the kernel only when its mathematical model is formalized, bounded, and verified against execution receipts. This monograph transitions all 115 capabilities from declared to admitted under the strict evidence bindings described in the appendices.

5 Evidence Method

To prevent "receipt theater"—where cryptographic language is used to mask unchecked execution—we enforce a rigid evidence method. Every capability claim in this monograph is bound to an actual implementation file, a specific dispatch symbol, and an execution log or test fixture.

The receipt status categorizes the strength of the evidence. A `VERIFIER_LOG_RECEIPTED` state implies that the execution was run through the automated verifier, producing a stdout log that captures the success or structural refusal of the test boundary. A `FIXTURE_EVIDENCED` state indicates that a fixture exists and passes unit testing, though a full verifier pipeline log may be pending.

Definition 5.1. Receipt Monoid. The set of all valid receipts $\mathcal{R}$ forms a monoid under the composition operator $\circ$, where $R_1 \circ R_2$ represents the cryptographic chaining of execution states.

This ensures that the entire trace of mechanical intelligence operations can be mathematically audited by a third party using only the binaries and the receipts.

6 Algebraic Model

The `wasm4pm` architecture maps algorithms to typed morphisms within specific algebraic structures. By modeling process graphs as idempotent semirings, we can formulate shortest-path algorithms (A^*) and reachability analysis using standard matrix multiplication over the semiring.

For temporal logic and constraints, we employ relation algebras and constraint lattices. The `allen_temporal` capability operates over a relation algebra where compositions of temporal intervals are evaluated via matrix intersections $R_{ik} \leftarrow R_{ik} \cap (R_{ij} \circ R_{jk})$.

Capabilities as Typed Morphisms. Every capability is a morphism mapping an input domain to a receipt codomain.

Boundary. These algebraic structures are explicitly bounded by the finite size of the trace window or model token limit. Infinite traces are structurally impossible in linear memory.

7 Discrete Geometry of Process and Cognition

We discard decorative geometric claims of "higher-dimensional temporal manifolds" in favor of bounded, discrete geometry. The event space of *wasm4pm* is modeled using trace metric spaces, process graph geometry, and constraint polytopes.

Trace metric spaces define the geometry for `compute_trace_similarity_matrix` and clustering algorithms. By defining a string edit distance or vector embedding, we can compute precise bounds on cluster radii.

Proposition 7.1. *The state-space geometry of the `transition_system` algorithm is a directed finite graph where the reachability set is strictly bounded by $O(|V| + |E|)$.*

For decision theory capabilities (`mdp`, `pomdp`), the geometry is the standard probability simplex. Operations map belief states from one point in the simplex to another, subject to discrete finite-horizon truncations to guarantee termination.

8 Discrete Calculus and Runtime Cost

Runtime behavior is evaluated not by unbounded performance claims, but by discrete calculus and cost decomposition. The latency of any operation T_{total} is decomposed as:

$$T_{\text{total}} = T_{\text{admit}} + T_{\mu, \text{bounded}} + T_{\text{receipt}} + T_{\text{emit}}$$

Micro-benchmarks for all capabilities have been executed against hardware profiles (e.g., using `bench-tools` or node benchmarks). The results empirically validate the hypothesis that $T_{\mu, \text{bounded}} \ll T_{\text{batch}} - T_{\mu}$, which justifies the first-mile architecture.

Continuous calculus is only used where computationally mapped to discrete differences, such as in `detect_drift` where $D_t = \|\theta_t - \theta_{t-1}\|$. All integrals are replaced with finite summations over the bounded event window.

9 Kernel Algorithms

The 60 kernel algorithms form the foundation of process intelligence. These are categorized into families: Discovery, Streaming, Analytics, Conformance, Simulation, Import/Export, OCEL, Prediction, ML, Social Network, and Agentic Pipelines.

Each algorithm family restricts the general execution environment. For example, Discovery algorithms like `dfg` and `alpha_miner` operate exclusively on finite matrices and sets. The transformation is bounded by the number of unique activities in the log.

Connection to Capability Families. The ML clustering algorithms leverage the trace metric spaces defined in Section 7. The exact mathematical mappings for all 60 algorithms are detailed in Appendix A, strictly enforcing the rule that no generic function signatures ($f : X \rightarrow Y$) are used.

10 Cognitive Breeds

The 55 cognitive breeds represent higher-order reasoning systems layered on top of the kernel algorithms. These include Temporal and Model Checking, Probabilistic Reasoning, Planning, SAT, and Constraint Logic.

Breeds like `ltl_monitor` and `sat_cdcl` do not claim to solve NP-hard problems instantaneously. Instead, they act as bounded monitors and solvers. If the resolution depth exceeds the heuristic limit, the system gracefully degrades, emitting a typed refusal receipt.

Proposition 10.1. *The `csp_ac3` breed strictly reduces the domain D_i in $O(ed^3)$ time. If arc consistency cannot be achieved within the linear memory boundary, it aborts rather than exhausting host resources.*

This bounding ensures that the cognitive layer remains stable at the edge.

11 Device-Side WASM Runtime

The Device-Side WASM Runtime is the execution engine enforcing these bounds. It operates entirely within a linear memory sanctuary. All state transitions within this domain are provably free from side effects of the host environment (no direct syscalls, no unmanaged entropy, no global clocks).

The WASM boundary acts as the ultimate security and performance governor. By stripping away OS-level multiplexing, the runtime ensures that capabilities execute with maximum cache locality.

Limitation. Because the runtime is perfectly deterministic, any capability requiring stochastic behavior (like Monte Carlo simulations) must be provided a deterministic seed via the input hash. True randomness is structurally prohibited.

12 Blue-Ocean Category: From ETL to First-Mile Admission

The *wasm4pm* project introduces a blue-ocean category by shifting process mining from post-hoc ETL analytics to synchronous, first-mile admission. The market value is not merely in accelerating existing algorithms, but in converting heuristic Python scripts into receipt-bearing, replayable, bounded execution objects.

In traditional systems, compliance violations are detected days after the event occurs. In the *wasm4pm* architecture, the violation is detected at the device edge, blocked or flagged, and cryptographically receipted.

Architectural Connection. This shift is fundamentally enabled by the Law-State framework (Section 4) and the strict WASM Runtime bounding (Section 11), ensuring that deploying logic to the edge does not compromise device stability.

13 Limits and Falsifiers

Mechanical intelligence is constrained by physical and computational realities. The system relies on bounded horizons.

Definition 13.1. Falsifiability of Execution. A capability claim is falsified if an input within the schema bounds produces an infinite loop, an out-of-memory panic, or a non-deterministic result on identical architectures.

Every capability in Appendices A and B specifies an item-specific falsifier. For example, `sat_cdcl` is falsified if it admits a known unsatisfiable formula. We reject "unbroken continuum of proof" as rhetorical overreach; the continuum is exactly as strong as the test fixtures and execution receipts currently binding it.

14 Conclusion

This monograph establishes the architectural, mathematical, and evidentiary foundation for *wasm4pm*. By elevating 115 capabilities from raw code to mathematically modeled, strictly receipted operators, we define the First-Mile Mechanical Intelligence framework.

The contribution of *wasm4pm* is not the claim that arbitrary unbounded reasoning problems become nanosecond-scale. The contribution is an architectural relocation: bounded, code-aligned process and cognitive operators are executed at the first mile of observation, inside a WASM runtime, so that raw events can be converted into admitted observations before batch transport.

Future work involves formal hardware benchmarking to convert our runtime calculus hypotheses into verified latency receipts. The current evidence graph stands as a rigorous, falsifiable baseline for autonomic process management.

A Algorithm Math Catalog

A.1 a_star

Implemented object. wasm4pm/src/fast_discovery.rs

Mathematical model. $f(n) = g(n) + h(n)$, $n^* = \arg \min_{n \in Open} f(n)$.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.1. Falsified if the selected path is not minimal under $f(n) = g(n) + h(n)$ for an admissible heuristic and bounded graph fixture.

A.2 aco

Implemented object. wasm4pm/src/genetic_discovery.rs

Mathematical model. $P_{ij}^{(k)} = \frac{\tau_{ij}^\alpha \eta_{ij}^\beta}{\sum_{\ell \in N_i^k} \tau_{i\ell}^\alpha \eta_{i\ell}^\beta}$, $\tau_{ij} \leftarrow (1 - \rho)\tau_{ij} + \Delta\tau_{ij}$.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.2. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.3 alpha_plus_plus

Implemented object. wasm4pm/src/algorithms.rs

Mathematical model. $T_{alphaplusplus}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.3. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.4 declare

Implemented object. wasm4pm/src/discovery.rs

Mathematical model. $T_{\text{declare}}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.4. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.5 dfg

Implemented object. wasm4pm/src/discovery.rs

Mathematical model. $W(a, b) = \sum_{\sigma \in L} \sum_{t=1}^{|\sigma|-1} \mathbf{1}[\sigma_t = a \wedge \sigma_{t+1} = b]$.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.5. Falsified if directly-follows counts do not equal adjacent activity-pair counts over the same trace log.

A.6 genetic_algorithm

Implemented object. wasm4pm/src/genetic_discovery.rs

Mathematical model. $P(x_i \text{ selected}) = \frac{F(x_i)}{\sum_j F(x_j)}$, $x' = \text{crossover}(x_a, x_b)$, $x'' = \text{mutate}(x')$.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.6. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.7 heuristic_miner

Implemented object. wasm4pm/src/advanced_algorithms.rs

Mathematical model. $\mathcal{G}_{\text{heuristicminer}} = (V, E)$, $E \subseteq V \times V$ bounded by event count.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.7. Falsified if the discovered structural graph fails to replay the exact log fixture under structural equivalence.

A.8 hill_climbing

Implemented object. wasm4pm/src/fast_discovery.rs

Mathematical model. $x_{t+1} = \arg \max_{x' \in N(x_t)} F(x')$.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.8. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.9 ilp

Implemented object. wasm4pm/src/ilp_discovery.rs

Mathematical model. $T_{ilp}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.9. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.10 inductive_miner

Implemented object. wasm4pm/src/more_discovery.rs

Mathematical model. $\mathcal{G}_{inductiveminer} = (V, E)$, $E \subseteq V \times V$ bounded by event count.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.10. Falsified if the discovered structural graph fails to replay the exact log fixture under structural equivalence.

A.11 optimized_dfg

Implemented object. wasm4pm/src/ilp_discovery.rs

Mathematical model. $T_{optimizeddfg}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.11. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.12 process_skeleton

Implemented object. wasm4pm/src/more_discovery.rs

Mathematical model. $T_{processskeleton}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.12. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.13 pso

Implemented object. wasm4pm/src/genetic_discovery.rs

Mathematical model. $v_{t+1} = \omega v_t + c_1 r_1(p_t - x_t) + c_2 r_2(g_t - x_t), \quad x_{t+1} = x_t + v_{t+1}.$

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.13. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.14 simulated_annealing

Implemented object. wasm4pm/src/more_discovery.rs

Mathematical model. $P(\text{accept}) = \min \left(1, \exp \frac{F(x') - F(x)}{T} \right).$

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.14. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.15 hierarchical_dfg

Implemented object. wasm4pm/src/hierarchical.rs

Mathematical model. $T_{\text{hierarchicaldfg}}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.15. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.16 simd_streaming_dfg

Implemented object. wasm4pm/src/simd_streaming_dfg.rs

Mathematical model. $S_{\text{simdstreamingdfg}}(t) = F(e_t, S(t-1))$ under strict Markov bound.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.16. Falsified if the windowed aggregation deviates from the strictly mapped continuous sum over the bounded stream fixture.

A.17 smart_engine

Implemented object. wasm4pm/src/smart_engine.rs

Mathematical model. $T_{\text{smartengine}}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.17. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.18 streaming_log

Implemented object. wasm4pm/src/streaming_wasm.rs

Mathematical model. $\mathcal{S}_{streaminglog}(t) = F(e_t, \mathcal{S}(t-1))$ under strict Markov bound.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.18. Falsified if the windowed aggregation deviates from the strictly mapped continuous sum over the bounded stream fixture.

A.19 analyze_process_speedup

Implemented object. wasm4pm/src/final_analytics.rs

Mathematical model. $T_{analyzeprocessspeedup}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.19. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.20 analyze_variant_complexity

Implemented object. wasm4pm/src/final_analytics.rs

Mathematical model. $T_{analyzevariantcomplexity}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.20. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.21 batches

Implemented object. `wasm4pm/src/batches.rs`

Mathematical model. $T_{batches}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `EXECUTION_RECEIPTED`

Falsifier A.21. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.22 causal_graph

Implemented object. `wasm4pm/src/causal_graph.rs`

Mathematical model. $T_{causalgraph}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `EXECUTION_RECEIPTED`

Falsifier A.22. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.23 compute_activity_transition_matrix

Implemented object. `wasm4pm/src/final_analytics.rs`

Mathematical model. $P_{ab} = \frac{W(a,b)}{\sum_{c \in \Sigma} W(a,c)}$.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `EXECUTION_RECEIPTED`

Falsifier A.23. Falsified if the numeric projection violates the known bounds of the fixture covariance or distance matrix.

A.24 compute_trace_similarity_matrix

Implemented object. `wasm4pm/src/final_analytics.rs`

Mathematical model. $S_{ij} = 1 - \frac{d(\sigma_i, \sigma_j)}{\max(|\sigma_i|, |\sigma_j|)}$.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.24. Falsified if the numeric projection violates the known bounds of the fixture covariance or distance matrix.

A.25 correlation_miner

Implemented object. wasm4pm/src/correlation_miner.rs

Mathematical model. $\mathcal{G}_{correlationminer} = (V, E)$, $E \subseteq V \times V$ bounded by event count.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.25. Falsified if the discovered structural graph fails to replay the exact log fixture under structural equivalence.

A.26 log_to_trie

Implemented object. wasm4pm/src/log_to_trie.rs

Mathematical model. $\text{insert}(\sigma) = v_0 \xrightarrow{\sigma_1} v_1 \xrightarrow{\sigma_2} \dots \xrightarrow{\sigma_k} v_k$.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.26. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.27 performance_spectrum

Implemented object. wasm4pm/src/performance_spectrum.rs

Mathematical model. $T_{performancespectrum}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.27. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.28 transition_system

Implemented object. wasm4pm/src/transition_system.rs

Mathematical model. $T_{transition_system}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.28. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.29 alignments

Implemented object. wasm4pm/src/alignments.rs

Mathematical model. $T_{alignments}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.29. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.30 complexity_metrics

Implemented object. wasm4pm/src/complexity_metrics.rs

Mathematical model. $T_{complexity_metrics}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.30. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.31 etconformance_precision

Implemented object. wasm4pm/src/etconformance_precision.rs

Mathematical model. $T_{etconformanceprecision}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIVED

Falsifier A.31. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.32 generalization

Implemented object. wasm4pm/src/generalization.rs

Mathematical model. $T_{generalization}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIVED

Falsifier A.32. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.33 monte_carlo_simulation

Implemented object. wasm4pm/src/montecarlo.rs

Mathematical model. $T_{montecarlosimulation}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIVED

Falsifier A.33. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.34 playout

Implemented object. wasm4pm/src/petri_net_playout.rs

Mathematical model. $T_{playout}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.34. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.35 bpmn_import

Implemented object. wasm4pm/src/bpmn_import.rs

Mathematical model. $T_{bpmnimport}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.35. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.36 pnml_import

Implemented object. wasm4pm/src/pnml_io.rs

Mathematical model. $\mathcal{A}_{pnmlimport} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ mapped via bounded tensor projection.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.36. Falsified if the numeric projection violates the known bounds of the fixture covariance or distance matrix.

A.37 powl_to_process_tree

Implemented object. wasm4pm/src/powl/conversion/to_process_tree.rs

Mathematical model. $\tau : \mathcal{L}_{\text{ext}} \rightarrow \mathcal{L}_{\text{kernel}}$ via deterministic AST traversal.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.37. Falsified if the parsed AST does not exhibit a bijective mapping to the valid internal structural schema.

A.38 yawl_export

Implemented object. wasm4pm/src/powl/conversion/to_yawl.rs

Mathematical model. $T_{yawlexport}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.38. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.39 ocel_dfg

Implemented object. wasm4pm/src/discovery.rs

Mathematical model. $\tau : \mathcal{L}_{\text{ext}} \rightarrow \mathcal{L}_{\text{kernel}}$ via deterministic AST traversal.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.39. Falsified if the parsed AST does not exhibit a bijective mapping to the valid internal structural schema.

A.40 ocel_dfg_per_type

Implemented object. wasm4pm/src/discovery.rs

Mathematical model. $\tau : \mathcal{L}_{\text{ext}} \rightarrow \mathcal{L}_{\text{kernel}}$ via deterministic AST traversal.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.40. Falsified if the parsed AST does not exhibit a bijective mapping to the valid internal structural schema.

A.41 ocel_encode

Implemented object. wasm4pm/src/text_encoding.rs

Mathematical model. $\tau : \mathcal{L}_{\text{ext}} \rightarrow \mathcal{L}_{\text{kernel}}$ via deterministic AST traversal.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.41. Falsified if the parsed AST does not exhibit a bijective mapping to the valid internal structural schema.

A.42 ocel_oc_declare

Implemented object. wasm4pm/src/advanced/oc_declare.rs

Mathematical model. $\tau : \mathcal{L}_{\text{ext}} \rightarrow \mathcal{L}_{\text{kernel}}$ via deterministic AST traversal.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.42. Falsified if the parsed AST does not exhibit a bijective mapping to the valid internal structural schema.

A.43 ocel_ocla

Implemented object. wasm4pm/src/advanced/ocla.rs

Mathematical model. $\tau : \mathcal{L}_{\text{ext}} \rightarrow \mathcal{L}_{\text{kernel}}$ via deterministic AST traversal.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.43. Falsified if the parsed AST does not exhibit a bijective mapping to the valid internal structural schema.

A.44 ocel_petri_net

Implemented object. wasm4pm/src/oc_petri_net.rs

Mathematical model. $\tau : \mathcal{L}_{\text{ext}} \rightarrow \mathcal{L}_{\text{kernel}}$ via deterministic AST traversal.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.44. Falsified if the parsed AST does not exhibit a bijective mapping to the valid internal structural schema.

A.45 compute_ewma

Implemented object. wasm4pm/src/prediction_drift.rs

Mathematical model. $z_t = \alpha x_t + (1 - \alpha)z_{t-1}$.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.45. Falsified if $z_t \neq \alpha x_t + (1 - \alpha)z_{t-1}$ on the fixture sequence.

A.46 detect_drift

Implemented object. wasm4pm/src/prediction_drift.rs

Mathematical model. $D_t = \|\theta_t - \theta_{t-1}\|$, $\text{drift}(t) = \mathbf{1}[D_t > \varepsilon]$.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.46. Falsified if the windowed aggregation deviates from the strictly mapped continuous sum over the bounded stream fixture.

A.47 predict_next_activity

Implemented object. wasm4pm/src/prediction.rs

Mathematical model. $\mathcal{A}_{\text{predictnextactivity}} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ mapped via bounded tensor projection.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.47. Falsified if the numeric projection violates the known bounds of the fixture covariance or distance matrix.

A.48 predict_outcome

Implemented object. wasm4pm/src/prediction_outcome.rs

Mathematical model. $\mathcal{A}_{predictoutcome} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ mapped via bounded tensor projection.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.48. Falsified if the numeric projection violates the known bounds of the fixture covariance or distance matrix.

A.49 predict_remaining_time

Implemented object. wasm4pm/src/prediction_remaining_time.rs

Mathematical model. $\mathcal{A}_{predictremainingtime} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ mapped via bounded tensor projection.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.49. Falsified if the numeric projection violates the known bounds of the fixture covariance or distance matrix.

A.50 automl_classify

Implemented object. wasm4pm/src/ml/automl.rs

Mathematical model. $\mathcal{A}_{automlclassify} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ mapped via bounded tensor projection.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.50. Falsified if the numeric projection violates the known bounds of the fixture covariance or distance matrix.

A.51 automl_forecast

Implemented object. wasm4pm/src/ml/automl.rs

Mathematical model. $\mathcal{A}_{automlforecast} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ mapped via bounded tensor projection.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.51. Falsified if the numeric projection violates the known bounds of the fixture covariance or distance matrix.

A.52 ml_anomaly

Implemented object. wasm4pm/src/anomaly.rs

Mathematical model. $\mathcal{A}_{mlanomaly} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ mapped via bounded tensor projection.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.52. Falsified if the numeric projection violates the known bounds of the fixture covariance or distance matrix.

A.53 ml_classify

Implemented object. wasm4pm/src/ml/classification.rs

Mathematical model. $\mathcal{A}_{mlclassify} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ mapped via bounded tensor projection.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.53. Falsified if the numeric projection violates the known bounds of the fixture covariance or distance matrix.

A.54 ml_cluster

Implemented object. wasm4pm/src/ml/clustering.rs

Mathematical model. $\arg \min_{C_1, \dots, C_k} \sum_{r=1}^k \sum_{x \in C_r} \|x - \mu_r\|^2.$

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.54. Falsified if the numeric projection violates the known bounds of the fixture covariance or distance matrix.

A.55 ml_forecast

Implemented object. wasm4pm/src/ml/forecasting.rs

Mathematical model. $\mathcal{A}_{mlforecast} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ mapped via bounded tensor projection.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.55. Falsified if the numeric projection violates the known bounds of the fixture covariance or distance matrix.

A.56 ml_pca

Implemented object. wasm4pm/src/ml/pca.rs

Mathematical model. $X_c = X - \mathbf{1}\mu^\top$, $\Sigma_X = \frac{1}{n-1}X_c^\top X_c$, $\Sigma_X v_i = \lambda_i v_i$.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier A.56. Falsified if the numeric projection violates the known bounds of the fixture covariance or distance matrix.

A.57 ml_regress

Implemented object. wasm4pm/src/ml/regression.rs

Mathematical model. $\hat{\beta} = (X^\top X)^{-1}X^\top y$.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIVED

Falsifier A.57. Falsified if the numeric projection violates the known bounds of the fixture covariance or distance matrix.

A.58 handover_network

Implemented object. wasm4pm/src/social_network.rs

Mathematical model. $T_{handovernetwork}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIVED

Falsifier A.58. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.59 working_together_network

Implemented object. wasm4pm/src/social_network.rs

Mathematical model. $T_{workingtogethernetwork}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIVED

Falsifier A.59. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

A.60 agentic_pipeline

Implemented object. wasm4pm/src/lib.rs

Mathematical model. $T_{agenticpipeline}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIVED

Falsifier A.60. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B Breed Math Catalog

B.1 ltl_monitor

Implemented object. `crates/wasm4pm-cognition/src/breeds/ltl_monitor.rs`

Mathematical model. $\sigma, t \models \mathbf{G}\varphi \iff \forall k \geq t, \sigma, k \models \varphi.$

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `TEST_LOG RECEIPTED`

Falsifier B.1. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.2 allen_temporal

Implemented object. `crates/wasm4pm-cognition/src/breeds/allen_temporal.rs`

Mathematical model. $R_{ik} \leftarrow R_{ik} \cap (R_{ij} \circ R_{jk}).$

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `TEST_LOG RECEIPTED`

Falsifier B.2. Falsified if A meets B and B during C does not reduce A:C to the expected composition mask.

B.3 ctl_check

Implemented object. `crates/wasm4pm-cognition/src/breeds/ctl_check.rs`

Mathematical model. $M, s \models \mathbf{EX}\varphi \iff \exists s'. (s, s') \in R \wedge M, s' \models \varphi.$

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `TEST_LOG RECEIPTED`

Falsifier B.3. Falsified if the inference resolution tree yields a contradiction on a known-satisfiable bounded hypothesis.

B.4 event_calculus

Implemented object. `crates/wasm4pm-cognition/src/breeds/event_calculus.rs`

Mathematical model. $\text{HoldsAt}(f, t) \leftarrow \exists e, t_0 < t : \text{Initiates}(e, f, t_0) \wedge \neg \text{Clipped}(t_0, f, t).$

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.4. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.5 situation_calculus

Implemented object. crates/wasm4pm-cognition/src/breeds/situation_calculus.rs

Mathematical model. $T_{\text{situationcalculus}}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.5. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.6 fuzzy_logic

Implemented object. crates/wasm4pm-cognition/src/breeds/fuzzy_logic.rs

Mathematical model. $\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x)), \quad \mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x)).$

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.6. Falsified if the inference resolution tree yields a contradiction on a known-satisfiable bounded hypothesis.

B.7 dempster_shafer

Implemented object. crates/wasm4pm-cognition/src/breeds/dempster_shafer.rs

Mathematical model. $m_{12}(A) = \frac{\sum_{B \cap C = A} m_1(B) m_2(C)}{1 - \sum_{B \cap C = \emptyset} m_1(B) m_2(C)}.$

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.7. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.8 abductive_ibe

Implemented object. crates/wasm4pm-cognition/src/breeds/abductive_ibe.rs

Mathematical model. $T_{abductiveibe}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.8. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.9 bayesian_network

Implemented object. crates/wasm4pm-cognition/src/breeds/bayesian_network.rs

Mathematical model. $P(X_1, \dots, X_n) = \prod_i P(X_i \mid Pa(X_i))$.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.9. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.10 problog

Implemented object. crates/wasm4pm-cognition/src/breeds/problog.rs

Mathematical model. $T_{problog}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.10. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.11 markov_logic

Implemented object. `crates/wasm4pm-cognition/src/breeds/markov_logic.rs`

Mathematical model. $\Phi \vdash_{\text{markovlogic}} \psi$ evaluated by bounded inference graph.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `TEST_LOG RECEIPTED`

Falsifier B.11. Falsified if the inference resolution tree yields a contradiction on a known-satisfiable bounded hypothesis.

B.12 htn_planning

Implemented object. `crates/wasm4pm-cognition/src/breeds/htn_planning.rs`

Mathematical model. $T_{\text{htnplanning}}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `TEST_LOG RECEIPTED`

Falsifier B.12. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.13 partial_order_plan

Implemented object. `crates/wasm4pm-cognition/src/breeds/partial_order_plan.rs`

Mathematical model. $T_{\text{partialorderplan}}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `FIXTURE_EVIDENCED`

Falsifier B.13. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.14 contingent_plan

Implemented object. `crates/wasm4pm-cognition/src/breeds/contingent_plan.rs`

Mathematical model. $T_{\text{contingentplan}}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `FIXTURE_EVIDENCED`

Falsifier B.14. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.15 mdp

Implemented object. `crates/wasm4pm-cognition/src/breeds/mdp.rs`

Mathematical model. $V_{k+1}(s) = \max_a [R(s, a) + \gamma \sum_{s'} P(s'|s, a) V_k(s')]$.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `FIXTURE_EVIDENCED`

Falsifier B.15. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.16 pomdp

Implemented object. `crates/wasm4pm-cognition/src/breeds/pomdp.rs`

Mathematical model. $b'(s') = \eta O(o|s', a) \sum_s P(s'|s, a) b(s)$.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `FIXTURE_EVIDENCED`

Falsifier B.16. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.17 strips

Implemented object. `crates/wasm4pm-cognition/src/breeds/strips.rs`

Mathematical model. $T_{strips}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. FIXTURE_EVIDENCED

Falsifier B.17. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.18 gps

Implemented object. crates/wasm4pm-cognition/src/breeds/gps.rs

Mathematical model. $T_{gps}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. FIXTURE_EVIDENCED

Falsifier B.18. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.19 asp

Implemented object. crates/wasm4pm-cognition/src/breeds/asp.rs

Mathematical model. $T_{asp}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. FIXTURE_EVIDENCED

Falsifier B.19. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.20 abductive_lp

Implemented object. crates/wasm4pm-cognition/src/breeds/abductive_lp.rs

Mathematical model. $T_{abductivelp}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. FIXTURE_EVIDENCED

Falsifier B.20. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.21 tableaux

Implemented object. `crates/wasm4pm-cognition/src/breeds/tableaux.rs`

Mathematical model. $T_{tableaux}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `FIXTURE_EVIDENCED`

Falsifier B.21. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.22 prolog

Implemented object. `crates/wasm4pm-cognition/src/breeds/prolog.rs`

Mathematical model. $T_{prolog}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `FIXTURE_EVIDENCED`

Falsifier B.22. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.23 clp

Implemented object. `crates/wasm4pm-cognition/src/breeds/clp.rs`

Mathematical model. $T_{clp}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `FIXTURE_EVIDENCED`

Falsifier B.23. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.24 sat_cdcl

Implemented object. `crates/wasm4pm-cognition/src/breeds/sat_cdcl.rs`

Mathematical model. $\varphi = \bigwedge_i \bigvee_j \ell_{ij}$.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `FIXTURE_EVIDENCED`

Falsifier B.24. Falsified if a known unsatisfiable CNF is admitted as satisfiable or a known satisfiable CNF is refused.

B.25 `csp_ac3`

Implemented object. `crates/wasm4pm-cognition/src/breeds/csp_ac3.rs`

Mathematical model. $D_i \leftarrow D_i \setminus \{x \in D_i : \nexists y \in D_j, C_{ij}(x, y)\}.$

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `TEST_LOG_RECEIPTED`

Falsifier B.25. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.26 `default_logic`

Implemented object. `crates/wasm4pm-cognition/src/breeds/default_logic.rs`

Mathematical model. $\Phi \vdash_{defaultlogic} \psi$ evaluated by bounded inference graph.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `TEST_LOG_RECEIPTED`

Falsifier B.26. Falsified if the inference resolution tree yields a contradiction on a known-satisfiable bounded hypothesis.

B.27 `circumscription`

Implemented object. `crates/wasm4pm-cognition/src/breeds/circumscription.rs`

Mathematical model. $T_{circumscription}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.27. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.28 frames_inheritance

Implemented object. crates/wasm4pm-cognition/src/breeds/frames_inheritance.rs

Mathematical model. $T_{framesinheritance}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.28. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.29 description_logic

Implemented object. crates/wasm4pm-cognition/src/breeds/description_logic.rs

Mathematical model. $\mathcal{I} \models C \sqsubseteq D \iff C^{\mathcal{I}} \subseteq D^{\mathcal{I}}$.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.29. Falsified if the inference resolution tree yields a contradiction on a known-satisfiable bounded hypothesis.

B.30 belief_merging

Implemented object. crates/wasm4pm-cognition/src/breeds/belief_merging.rs

Mathematical model. $T_{beliefmerging}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.30. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.31 script_sam

Implemented object. `crates/wasm4pm-cognition/src/breeds/script_sam.rs`

Mathematical model. $T_{scriptsam}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.31. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.32 act_r

Implemented object. `crates/wasm4pm-cognition/src/breeds/act_r.rs`

Mathematical model. $T_{actr}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.32. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.33 soar

Implemented object. `crates/wasm4pm-cognition/src/breeds/soar.rs`

Mathematical model. $T_{soar}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.33. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.34 episodic_memory

Implemented object. `crates/wasm4pm-cognition/src/breeds/episodic_memory.rs`

Mathematical model. $T_{episodicmemory}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG_RECEIPTED

Falsifier B.34. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.35 ebl

Implemented object. crates/wasm4pm-cognition/src/breeds/ebl.rs

Mathematical model. $T_{ebl}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG_RECEIPTED

Falsifier B.35. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.36 ilp

Implemented object. crates/wasm4pm-cognition/src/breeds/ilp.rs

Mathematical model. $T_{ilp}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. EXECUTION_RECEIPTED

Falsifier B.36. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.37 version_space

Implemented object. crates/wasm4pm-cognition/src/breeds/version_space.rs

Mathematical model. $T_{version_space}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.37. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.38 analogy_sme

Implemented object. crates/wasm4pm-cognition/src/breeds/analogy_sme.rs

Mathematical model. $T_{analogy_sme}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.38. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.39 rl_symbolic

Implemented object. crates/wasm4pm-cognition/src/breeds/rl_symbolic.rs

Mathematical model. $Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.39. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.40 qualitative_reason

Implemented object. crates/wasm4pm-cognition/src/breeds/qualitative_reason.rs

Mathematical model. $T_{qualitative_reason}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.40. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.41 naive_physics

Implemented object. `crates/wasm4pm-cognition/src/breeds/naive_physics.rs`

Mathematical model. $T_{naivephysics}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `TEST_LOG RECEIPTED`

Falsifier B.41. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.42 triz

Implemented object. `crates/wasm4pm-cognition/src/breeds/triz.rs`

Mathematical model. $T_{triz}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `TEST_LOG RECEIPTED`

Falsifier B.42. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.43 morphological

Implemented object. `crates/wasm4pm-cognition/src/breeds/morphological.rs`

Mathematical model. $\Phi \vdash_{morphological} \psi$ evaluated by bounded inference graph.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `TEST_LOG RECEIPTED`

Falsifier B.43. Falsified if the inference resolution tree yields a contradiction on a known-satisfiable bounded hypothesis.

B.44 construction_grammar

Implemented object. `crates/wasm4pm-cognition/src/breeds/construction_grammar.rs`

Mathematical model. $T_{constructiongrammar}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG_RECEIPTED

Falsifier B.44. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.45 meta_reasoning

Implemented object. crates/wasm4pm-cognition/src/breeds/meta_reasoning.rs

Mathematical model. $T_{metareasoning}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG_RECEIPTED

Falsifier B.45. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.46 autoinstinct_learning

Implemented object. crates/wasm4pm-cognition/src/breeds/autoinstinct_learning.rs

Mathematical model. $T_{autoinstinctlearning}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG_RECEIPTED

Falsifier B.46. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.47 autoinstinct_neurosis

Implemented object. crates/wasm4pm-cognition/src/breeds/autoinstinct_neurosis.rs

Mathematical model. $T_{autoinstinctneurosis}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG_RECEIVED

Falsifier B.47. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.48 autoinstinct_semantics

Implemented object. crates/wasm4pm-cognition/src/breeds/autoinstinct_semantics.rs

Mathematical model. $T_{autoinstinctsemantics}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG_RECEIVED

Falsifier B.48. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.49 autoinstinct_vision

Implemented object. crates/wasm4pm-cognition/src/breeds/autoinstinct_vision.rs

Mathematical model. $T_{autoinstinctvision}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG_RECEIVED

Falsifier B.49. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.50 cbr

Implemented object. crates/wasm4pm-cognition/src/breeds/cbr.rs

Mathematical model. $T_{cbr}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG_RECEIVED

Falsifier B.50. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.51 dendral

Implemented object. `crates/wasm4pm-cognition/src/breeds/dendral.rs`

Mathematical model. $T_{dendral}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `TEST_LOG RECEIPTED`

Falsifier B.51. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.52 eliza

Implemented object. `crates/wasm4pm-cognition/src/breeds/frame.rs`

Mathematical model. $T_{eliza}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `TEST_LOG RECEIPTED`

Falsifier B.52. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.53 hearsay

Implemented object. `crates/wasm4pm-cognition/src/breeds/hearsay.rs`

Mathematical model. $T_{hearsay}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. `TEST_LOG RECEIPTED`

Falsifier B.53. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.54 mycin

Implemented object. `crates/wasm4pm-cognition/src/breeds/production_rules.rs`

Mathematical model. $T_{mycin}(S) \subseteq S \times S$ representing safe state transitions.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.54. Falsified if the generated artifact fails to map deterministically to the known receipt hash for the fixture input.

B.55 ocpm_route_discoverer

Implemented object. crates/wasm4pm-cognition/src/breeds/ocpm_route_discoverer.rs

Mathematical model. $\mathcal{G}_{ocpmroutediscoverer} = (V, E)$, $E \subseteq V \times V$ bounded by event count.

Bounded execution claim. Bounded to finite structure traversal. Assumes bounded input matrix or token graph.

Receipt status. TEST_LOG RECEIPTED

Falsifier B.55. Falsified if the discovered structural graph fails to replay the exact log fixture under structural equivalence.

C Receipt Index

Capability	State	Math Model Status
a_star	EXECUTION_RECEIPTED	CODE_DERIVED_FORMULA
aco	EXECUTION_RECEIPTED	CODE_DERIVED_FORMULA
alpha_plus_plus	EXECUTION_RECEIPTED	STANDARD_FORMULA_WIT
declare	EXECUTION_RECEIPTED	STANDARD_FORMULA_WIT
dfg	EXECUTION_RECEIPTED	CODE_DERIVED_FORMULA
genetic_algorithm	EXECUTION_RECEIPTED	CODE_DERIVED_FORMULA
heuristic_miner	EXECUTION_RECEIPTED	STANDARD_FORMULA_WIT
hill_climbing	EXECUTION_RECEIPTED	CODE_DERIVED_FORMULA
ilp	EXECUTION_RECEIPTED	STANDARD_FORMULA_WIT
inductive_miner	EXECUTION_RECEIPTED	STANDARD_FORMULA_WIT
optimized_dfg	EXECUTION_RECEIPTED	STANDARD_FORMULA_WIT
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detect_drift	EXECUTION_RECEIPTED	CODE_DERIVED_FORMULA
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ctl_check	TEST_LOG_RECEIPTED	CODE_DERIVED_FORMULA
event_calculus	TEST_LOG_RECEIPTED	CODE_DERIVED_FORMULA
situation_calculus	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
fuzzy_logic	TEST_LOG_RECEIPTED	CODE_DERIVED_FORMULA
dempster_shafer	TEST_LOG_RECEIPTED	CODE_DERIVED_FORMULA
abductive_ibe	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
bayesian_network	TEST_LOG_RECEIPTED	CODE_DERIVED_FORMULA
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contingent_plan	FIXTURE_EVIDENCED	STANDARD_FORMULA_WIT

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pomdp	FIXTURE_EVIDENCED	CODE_DERIVED_FORMULA
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tableaux	FIXTURE_EVIDENCED	STANDARD_FORMULA_WIT
prolog	FIXTURE_EVIDENCED	STANDARD_FORMULA_WIT
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description_logic	TEST_LOG_RECEIPTED	CODE_DERIVED_FORMULA
belief_merging	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
script_sam	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
act_r	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
soar	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
episodic_memory	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
eb1	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
ilp	EXECUTION_RECEIPTED	STANDARD_FORMULA_WIT
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analogy_sme	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
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morphological	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
construction_grammar	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
meta_reasoning	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
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autoinstinct_neurosis	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
autoinstinct_semantics	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
autoinstinct_vision	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
cbr	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
dendral	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
eliza	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
hearsay	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
mycin	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
ocpm_route_discoverer	TEST_LOG_RECEIPTED	STANDARD_FORMULA_WIT
